

NOTNT

SomeonecoolLovesMaths

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Problem

Let m and n be positive integers. Find the number of triplets of non-negative integers (x, y, z) that satisfy

$$mx + ny + z = mn.$$

Solution. It is enough to find the number of pairs of non-negative integers such that $mx + ny \leq mn$ since we can just let $z = mn - mx - ny \geq 0$.

Observe that this is the number of lattice points that either lie on the axes or in the first quadrant but not above that line $mx + ny = mn$. The line is basically the line passing through $(n, 0)$ and $(0, m)$.

Consider the rectangle of the points $(0, 0), (n, 0), (0, m), (n, m)$. This rectangle has a total of $(m + 1)(n + 1)$ points. Since we have half as many points in the rectangle, we have $\frac{(m + 1)(n + 1)}{2}$. In this case, we are undercounting the points that lie on the line $mx + ny = mn$ since they are only getting counted once.

Let $d = \gcd(m, n)$ and $m = da, n = db$. Then we have that $ax + by = dab$. Since $a \mid by \implies a \mid y$. So $y = ak_1$ and similarly $x = bk_2$. So we have $k_1 + k_2 = d$ where $0 \leq k_1, k_2 \leq d$. There are $d + 1$ such pairs of (k_1, k_2) , so we have $d + 1$ values of (x, y) which we have not counted.

As a result, the total number of points are $\frac{(m + 1)(n + 1) + \gcd(m, n) + 1}{2}$.